LUNARA

A Postpartum Support Platform

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Grand Canyon University • College of Engineering & Technology • CST-452 Senior Capstone

lunaracare.org

PROJECT DESCRIPTION & JUSTIFICATION

Project Description

LUNARA is a full-stack web application designed to support postpartum care by connecting mothers with certified doula providers through a unified digital platform. The system integrates core features that streamline care delivery and improve communication, including:

- Structured intake assessments for personalized onboarding
- Daily mood and symptom tracking with clinical alert detection
- Real-time messaging between clients and providers
- Appointment scheduling and automated reminders
- Personalized care plans with milestone tracking
- A curated educational resource library

Role-based access control ensures tailored experiences for Clients, Providers, and Administrators, supporting a scalable and secure system architecture.

Project Justification

Approximately 1 in 5 mothers experience a perinatal mood or anxiety disorder, yet fewer than 15% receive adequate treatment (Maternal Mental Health Leadership Alliance, 2023). The postpartum period introduces significant barriers to care, including:

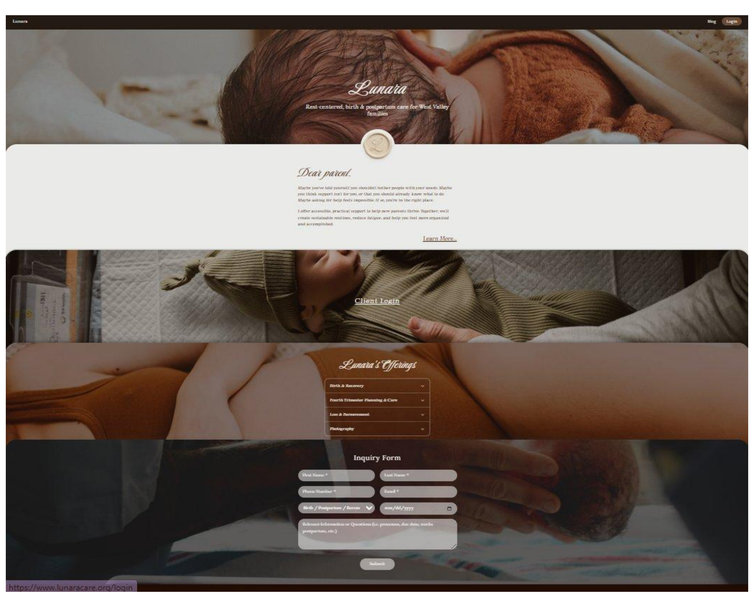
- Physical recovery and ongoing health challenges
- Sleep deprivation and increased stress levels
- Fragmented communication across care providers
- Limited access to centralized postpartum support resources

Although doula support has been shown to improve maternal outcomes, the process of finding, coordinating, and maintaining care remains largely manual and inaccessible.

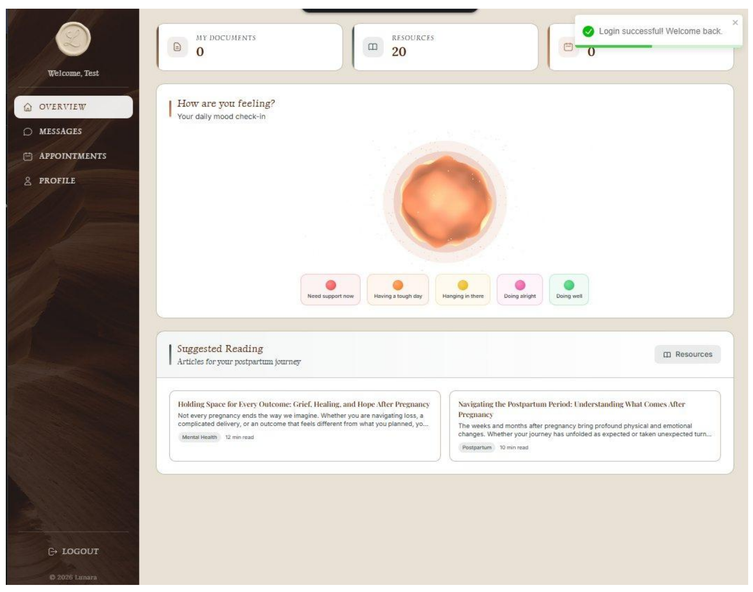
LUNARA addresses this gap by centralizing the postpartum care workflow into a single, integrated platform. Key benefits include:

- Reduced fragmentation through unified communication and scheduling
- Improved continuity of care between clients and providers
- Early detection of mental health risks through clinical alert monitoring
- Increased accessibility to postpartum resources and support systems

By combining health tracking, communication, and scheduling into one system, LUNARA enables more proactive and data-driven care, helping providers identify risks earlier and intervene more effectively.



Landing Page — Public-Facing Website (FR1)



Client Dashboard — Check-In & Resource Access (FR3, FR8)

FUNCTIONAL REQUIREMENTS & STANDARDS

The platform was developed against 15 functional requirements defined in the CST-451 Requirements Document. Eleven core features were successfully implemented across three iterations, delivering a fully integrated, real-time postpartum care platform with secure authentication, clinical monitoring, and provider–client interaction.

ID	Requirement	Status
FR1	Public website with inquiry form	✓
FR2	Authentication & registration (JWT, OAuth, MFA)	✓
FR3	Client & provider dashboards	✓
FR4	Dynamic five-step intake forms	✓
FR5	Real-time messaging (Socket.IO)	✓
FR6	Appointment scheduling & reminders	✓
FR7	Educational resource library	✓
FR8	Daily check-ins with clinical alerts	✓
FR9	Provider client management	✓
FR10	Care plan system with milestones	✓
FR11	Blog platform with draft/publish	✓
FR12	Digital journaling	Deferred
FR13	Daily insights	Deferred
FR14	Sleep & feeding trackers	Deferred
FR15	AI-powered note summarization	Deferred

Standards: OWASP Top 10 security compliance, WCAG 2.1 AA accessibility, RESTful API design, Agile sprint-based delivery.

DESIGN PROCESS

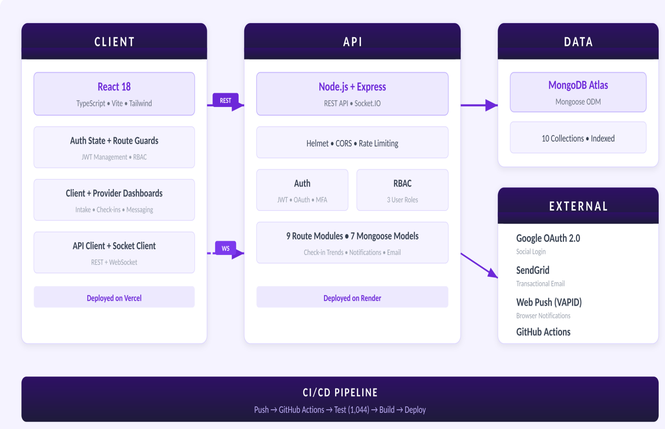
Development followed an iterative, sprint-based approach across three coding milestones using Agile methodology. Each iteration progressed through a structured workflow:

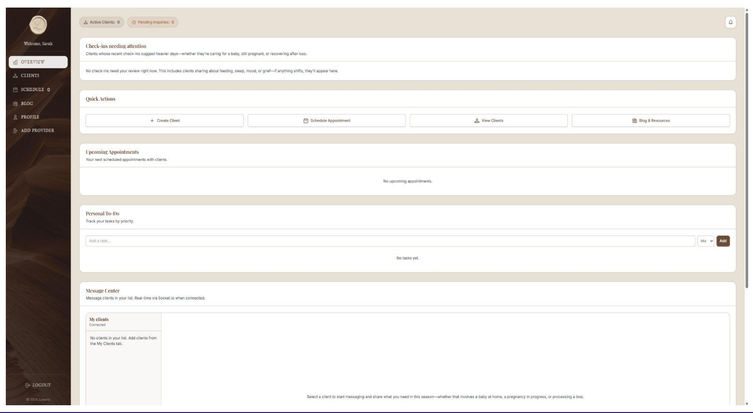
- Define scoped user stories and objectives
- Implement backend routes, services, and data models
- Develop frontend components and user interfaces
- Integrate and validate features in a live environment
- Refine based on testing, stability, and usability feedback

Project planning and task management were coordinated using GitHub Project Boards. All code changes were submitted through a pull request workflow with peer review to ensure code quality and consistency.

A GitHub Actions CI/CD pipeline automated test execution on every push, enforcing quality gates prior to deployment. The application was deployed using modern cloud infrastructure, with the backend hosted on Render, the frontend on Vercel, and MongoDB Atlas serving as the managed database layer.

SYSTEM ARCHITECTURE





DATA SUMMARY: TESTING & QUALITY

1,044	100%	118
Automated Tests	Pass Rate	Test Suites

90.58%	63.35%
Backend Statement Coverage	Frontend Statement Coverage

Test Category	Suites	Tests
Backend Unit	5	96
Backend Integration	7	57
Backend Services	8	57
Frontend Components	98	715
E2E (Playwright)	—	32
Manual Test Cases	—	20

Zero open defects at submission. Full suite executes in 25.6 seconds. CI/CD pipeline enforces all quality gates on every push.

CHALLENGES & SOLUTIONS

Team Sequencing
Andrew completed Capstone II before Carter and Owen, requiring him to define the initial system and determine a workable scope independently.
Solution: He built a modular backend foundation with clear documentation so the team could extend the system without major refactoring.

Communication in an Online Environment
Working remotely made coordination and alignment more difficult across development tasks.
Solution: The team used regular check-ins, GitHub project boards, and pull request reviews to stay aligned and maintain consistency.

System Complexity and Integration
Integrating multiple features such as authentication, real time messaging, and dashboards introduced complexity across the full stack.
Solution: A layered architecture and service based design helped separate concerns and simplify integration.

Real Time Communication
Implementing reliable real time messaging between users required synchronization between frontend and backend systems.
Solution: Socket.IO with JWT authentication was used to enable secure and consistent real time communication.

PROJECT SCOPE & COMPLETION

The project requirements included 15 functional features. Across three development iterations, the team successfully delivered **11 core features**, achieving **73% overall project completion**. The remaining four features include digital journaling, daily insights, sleep and feeding trackers, and AI powered summarization. These were intentionally deferred after evaluation.

These features required more advanced integration and additional development time beyond the project timeline. As a result, the team prioritized completing all core system functionality to ensure a stable and fully integrated application.

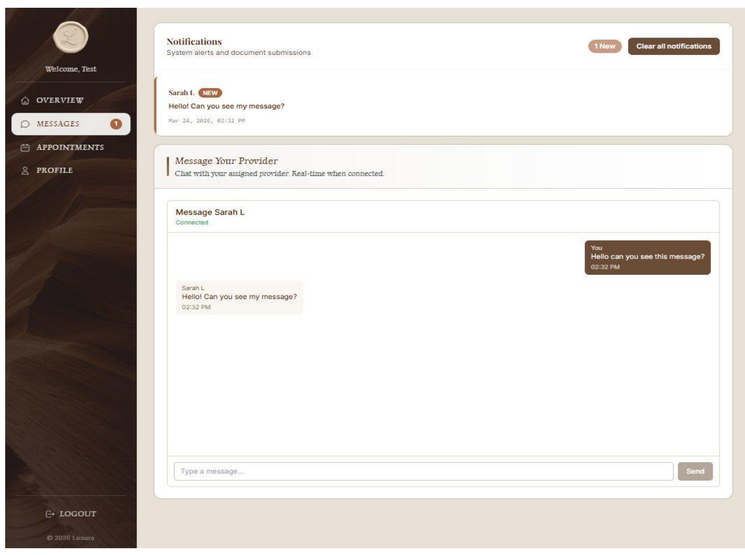
The system's modular architecture allows these deferred features to be implemented after launch without requiring major structural changes. This supports future scalability and continued development.

All core platform functionality was completed, with remaining features for post launch expansion.

FUTURE WORK

Near term priorities focus on validating the platform in real world environments through beta user recruitment and refining clinical workflows with active doula and client pairs. Payment processing integration will also be implemented to support production readiness.

Long term development will expand platform capabilities with a Progressive Web App for mobile access, AI powered session note summarization, and additional health tracking modules such as sleep and feeding trackers.



Real-Time Messaging — Client–Provider Chat (FR5)

VISIT THE SITE

Scan to explore the live platform

lunaracare.org

github.com/omniV1/AQC

Source Code Repository

Owen Lindsey Full-Stack Developer

Carter Wright Full-Stack Developer

Andrew Mack Full-Stack Developer

Advisor: Professor Amr Elchouemi

CST-452: Senior Capstone Project II